

VISION-ASSISTED DUAL-ARM CABLE HANDOVER AND INSERTION

Major Project Portfolio | FANUC India – Robot Department Internship Application

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Project focus: Autonomous dual-arm manipulation for cable handover and connector insertion using vision-based pose estimation, ROS 2 and MoveIt 2.

Current status: In development — CAD and URDF/XACRO work completed/in progress, followed by ROS 2, MoveIt 2, vision integration and validation.

1. Problem & Motivation

Cable and connector insertion is a challenging manipulation task because the cable is flexible, its shape and tip orientation can change during handling, and the connector requires precise alignment. A single arm cannot simultaneously stabilise the cable and perform insertion, motivating coordinated dual-arm manipulation.

The two arms share a workspace, so the system must avoid inter-arm, arm-cable and arm-environment collisions during pick, handover and insertion.

2. Project Objective

Develop an autonomous dual-arm system that detects cable and connector poses using fiducial (ArUco) markers, plans a collision-free handover in MoveIt 2, and performs compliant insertion. The system is intended to be validated in simulation and on low-cost real hardware.

Key objectives

- ArUco marker-based 3D pose estimation for cable and connector components.
- MoveIt 2-based dual-arm coordination and manipulation.
- Collision-free trajectory planning in a shared workspace.
- Validation through cable insertion and peg-in-hole tasks, with evaluation of reliability, positioning accuracy and execution time.

3. Robot Design & CAD

The project uses a low-cost 7-DoF bio-inspired arm concept driven by ST3215 serial-bus servos. The prototype was redesigned after the first iteration to improve structural rigidity, workspace, precision, actuator integration and compatibility with ROS 2 and MoveIt 2.

The CAD phase includes mechanical modelling of the arm, actuator integration and preparation of the robot model for URDF/XACRO conversion and subsequent simulation.

Design progression

Stage	Work
Initial prototype	Basic concept demonstrated; mechanical and integration issues identified.
Redesign	Improved rigidity, workspace, precision and actuator integration.
Robot description	CAD model being converted into URDF/XACRO for ROS 2.
Next stage	MoveIt 2 configuration, planning groups, kinematics, controllers and simulation.

4. Robotics Software & Motion Planning

The robot is being modelled in ROS 2 using URDF/XACRO and prepared for MoveIt 2-based planning. MoveIt 2 will be configured for left-arm, right-arm and dual-arm planning groups, end effectors, virtual joints, collision matrices, kinematics and controllers.

Motion-planning workflow

- Robot model → URDF/XACRO
- ROS 2 robot description and visualization
- MoveIt 2 configuration
- Target pose from RViz or the vision system
- Inverse kinematics to determine joint configurations
- OMPL-based trajectory generation
- Collision checking against the other arm, cable, table and environment
- RViz visualization before execution

Vision: ArUco markers are planned for pose estimation of the cable and connector, enabling the manipulation system to move from manually specified targets toward vision-guided autonomous operation.

5. Current Progress & My Contribution

The project is **currently in development**. The project timeline places CAD completion and URDF/XACRO conversion in the current phase, followed by ROS 2/MoveIt 2 modelling and visualization, simulation and control, motion planning, vision integration, cable-insertion logic, and final testing.

My role

- Mechanical/CAD modelling of the robotic arm and prototype development.
- URDF/XACRO robot-description preparation.
- ROS 2 integration and preparation for MoveIt 2 simulation.
- Support for motion-planning and dual-arm manipulation development.

Planned validation includes autonomous peg-in-hole and cable-insertion tasks, followed by performance evaluation and testing on the real robotic platform.

Technologies

ROS 2 • MoveIt 2 • Python • OpenCV • ArUco • URDF/XACRO • RViz • Gazebo • Motion Planning • CAD • Serial-bus Servos

6. Relevant Achievement

My previous robotics project, **SAM – Multilingual AI-Based Educational Robot**, was awarded **Best Project of the Year 2025–2026** at the 49th Series Student Project Programme (SPP), organized by the Karnataka State Council for Science and Technology (KSCST). The project was recognized at the state-level presentation and exhibition.

